

Operating state up to t

wind and photovoltaic
measurements, forecasts
and 10-90% bands;
imbalance, ACE, prices
(Elia open data, 15 min)

AERIS

temporal encoder and
variational risk state
of the last T intervals
→ risk score s_t for
the interval $t+h$

Ranked alert list

intervals ordered by s_t ;
the top 2% are flagged
(one to two per day),
each with a density
confidence c_t

Operator action

15 to 60 min ahead:
reposition reserve,
schedule flexible units,
activate demand
response

